

Station name	QAS L
People present	JEB, SS, Armin
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	1 Sep 2021

Notes:

1560 cm

120 cm existing hydraulic fluid height

122 + 600 - 10 = 5250 height

AWS to do list		Check after maintenance	
Part	Part shipped (Y/N)	Shipped part number	Part number upon departure.
Radiometer <i>before on station 171515</i>			212891
Ablation hose assembly			Clean? ☒ Y/N
Iridium modem IMEI number			5002
Inclinometer/compass			No air in hose and bladder half full? Y/N
Satellite antenna			Attached to the radiometer? ☒ Y/N
Wind sensor			Black box pointing toward the radiometer (i.e. along the boom)? Y/N
Temperature / humidity assembly			Fan spins? Y/N Sounds OK? ☒ Y/N
HygroClip in T/hum assembly			NB: could have a 40 °C offset old: 61459360 new: 61180276
Sonic ranger on AWS			Old sensor with new membrane? ☒ Y/N
Sonic ranger on stake(s)			Old sensor with new membrane? Y/N NB: make sure not being buried next winter replaced both
Thermistor string			existing has 4.35 m to top sensor 3/15/29
Solar panel			Clean? ☒ Y/N
GPS antenna			NB: is not effective when partially covered
Data logger			At top of logger enclosure? ☒ Y/N
Multiplexer			CR1000 or CR1000X?
Logger enclosure			Clicking during measurement cycle? Y/N
Battery box			Replaced (including everything inside)? ☒ Y/N
Stakes			New desiccant bags? ☒ Y/N
			New box? ☒ Y/N
			New desiccant bags? ☒ Y/N
			New one(s) drilled in? ☒ Y/N
			Length of new stake(s) 8 m? Y/N
			Bottom plugged to reduce melt-in? ☒ Y/N
			New tripod? ☒ Y/N
			New sensor boom? ☒ Y/N

DWIAT

± 628 cm away from vertical
downslope + 160° rotation

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	2 sec	
Download data to PC and/or change CF Card (wait for green light)	Y/N	Y/N
Name of logger program	Proimize 2015e	
Screen dump of values in fast scan mode	Y/N	Y/N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	°
Mast tilt across boom looking from radiometer (- if clockwise)	°	°
Boom direction relative to north	used app 0 ° true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y/N	Y/N
Radiometer exactly aligned with inclinometer? If not, measure.	Y/N	Y/N
Temperature / humidity height (bottom of casing – surface)	238 cm	235 cm
Stake assembly standing?	see notes other side Y/N / sort of	Y/N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	" cm	cm
Free length of stake 2 (if 3: middle)	N/A cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	N/A cm	cm
Height of ablation bladder fluid level above the ice surface	120 cm	159 cm
Vertical difference in surface height (at station – at hose) (approx.)	+10 cm	+15* cm
Length of thermistor string on surface from surface marking	cm	cm
Apparent damage to sensors? If yes, which?	Y/N	Y/N

Notes:

rain hobo bucked fell down *
* hole for PTA is lower

Must-do at next visit:

use luftt rain gauge,
i.e. v3 station

Station name	QAS m
People present	JEB, JJ, pilot Sam
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	31 Aug 2021

Notes:

new GNSS install,
swapped Hobo rain
propeller was very loose, tightened

AWS to do list		Check after maintenance	
Part	Part shipped (Y/N)	Shipped part number	Part number upon departure.
Radiometer		171514	212889
Ablation hose assembly		010	
Iridium modem IMEI number			
Inclinometer/compass		Attached to the radiometer? ☒ Y/N	
Satellite antenna		✓	
Wind sensor		Black box pointing toward the radiometer (i.e. along the boom) ☒ Y/N	
Temperature / humidity assembly		Fan spins? ☒ Y/N Sounds OK? ☒ Y/N	
HygroClip in T/hum assembly		NB: could have a 40 °C offset	
Sonic ranger on AWS		Old sensor with new membrane? Y/N	
Sonic ranger on stake(s)		Old sensor with new membrane? Y/N NB: make sure not being buried next winter	
Thermistor string		none installed, existing was out by 5.42 m	
Solar panel		Clean? ☒ Y/N NB: is not effective when partially covered	
GPS antenna		At top of logger enclosure? Y/N	
Data logger		☒ CR1000 or CR1000X?	
Multiplexer		Clicking during measurement cycle? ☒ Y/N	
Logger enclosure		Replaced (including everything inside)? ☒ Y/N New desiccant bags? ☒ Y/N	
Battery box		New box? Y/N New desiccant bags? Y/N	
Stakes		New one(s) drilled in? ☒ Y/N Length of new stake(s) 8 m? ☒ Y/N Bottom plugged to reduce melt-in? ☒ Y/N	
Tripod		New tripod? ☒ Y/N New sensor boom? ☒ Y/N	

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		
Download data to PC and/or change CF Card (wait for green light)	☒ Y / N	Y / N
Name of logger program	PROMICE 2015 E	
Screen dump of values in fast scan mode	Y / N	Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	☒ Y / N	Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	+1 °	+1 °
Mast tilt across boom looking from radiometer (- if clockwise)	°	-2 °
Boom direction relative to north	55° CW from N true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	☒ Y / N	☒ Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	☒ Y / N	☒ Y / N
Temperature / humidity height (bottom of casing - surface)	cm	2.55 cm
Stake assembly standing?	☒ Y / N / sort of	Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	186 cm	186! cm
Vertical difference in surface height (at station - at hose) (approx.)	-25 cm	-10 cm
Length of thermistor string on surface from surface marking	cm	4.25 cm
Apparent damage to sensors? If yes, which?	see wind notes ☒ Y / N	☒ Y / N

Notes:

carbon boom 1.42 + 2.00 - 0.1
 S↓ 190 , S↑ 40 → to alu junction
 L↓ 14 , L↑ -70 , T 4.8C
 solar charge current WAN
 3.3 m surface height
 ice height 22.9

Must-do at next visit: snow fox wind, vertical axis

Station name	QAS-U V2 AWS
People present	JEB, JS, Armin Doehner, Pilot Sam
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	1 Sep, 2 Sep

Notes:

AWS V2 + V3 + SR50 carbon booms all bent down by settling snow

V2 AWS in ~25 degrees / wet hole, shallow ~30cm deep 5m wide

AWS to do list		Check after maintenance	
Part	Part shipped (Y/N)	Shipped part number	Part number upon departure.
Radiometer			030710 Clean? ☒ Y/N same instr as existing
Ablation hose assembly		did not change 3180	No air in hose and bladder half full? ☒ Y/N
Iridium modem IMEI number			300234061470510
Inclinometer/compass		Attached to the radiometer? ☒ Y/N	
Satellite antenna		sam	
Wind sensor		Black box pointing toward the radiometer (i.e. along the boom)? ☒ Y/N	
Temperature / humidity assembly		Fan spins? ☒ Y/N Sounds OK? ☒ Y/N	
HygroClip in T/hum assembly	height before maint 2.23	NB: could have a 40 °C offset 2.45 after move AWS to	
Sonic ranger on AWS		Old sensor with new membrane? ☒ Y/N changed flat across *	
Sonic ranger on stake(s)		Old sensor with new membrane? ☒ Y/N NB: make sure not being buried next winter changed	
Thermistor string		-8cm on 4th from top	
Solar panel		Clean? ☒ Y/N NB: is not effective when partially covered * *	
GPS antenna		At top of logger enclosure? ☒ Y/N	
Data logger		CR1000 or CR1000X?	
Multiplexer		Clicking during measurement cycle? ☒ Y/N	
Logger enclosure		Replaced (including everything inside)? ☒ Y/N New desiccant bags? ☒ Y/N	
Battery box		New box? ☒ Y/N New desiccant bags? ☒ Y/N	
Stakes V2 height 2.01 m		New one(s) drilled in? ☒ Y/N Length of new stake(s) 8 m? ☒ Y/N 4 m in ice, 4m high	
V3 2.09 m		Bottom plugged to reduce melt-in? ☒ Y/N	
Tripod		New tripod? ☒ Y/N replaced bent leg! :)	
		New sensor boom? ☒ Y/N	

re-installed Hobo rain

* shifted Rotronic downward
 ** " solar upward, 2.04 m bottom

Metadata maintenance

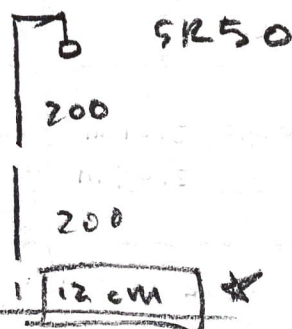
	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		+ 44 sec
Download data to PC and/or change CF Card (wait for green light)	Y / N	Y / N
Name of logger program		
Screen dump of values in fast scan mode	☒ Y / N	☒ Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N	Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	level °
Mast tilt across boom looking from radiometer (- if clockwise)	°	level °
Boom direction relative to north	° true/magn.	14° E ° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y / N	of South Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	☒ Y / N	☒ Y / N
Temperature / humidity height (bottom of casing – surface)	cm	245 cm
Stake assembly standing?	☒ Y / N / sort of	☒ Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	400 cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	cm	153 cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	-3 cm
Length of thermistor string on surface from surface marking	cm	cm
Apparent damage to sensors? If yes, which?	Y / N	H ☒ Y / N

Notes:

CF card question if data anyway gathered from logger +
 Hygroclip is the new type, unchanged
 ++ rain gauge, Hobo is attached in a work around way

Must-do at next visit:

taller masts!



Station name	RASU V3
People present	JEB, JJ, Armin, Schmieg, Sarah
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	2 Sep 2021

Notes:

clinometer removed for debug in camp
battery VDC 13.98

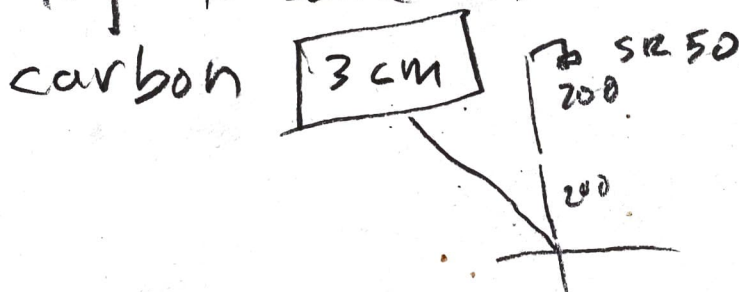
AWS to do list		Check after maintenance		
Part	Part shipped (Y/N)	Shipped part number	Part number upon departure.	
Radiometer			?	Clean? ☒ Y/N
Ablation hose assembly			none!	No air in hose and bladder half full? Y/N
Iridium modem IMEI number				4530 last 4
Inclinometer/compass		Attached to the radiometer? ☒ Y/N removed see above		
Satellite antenna				
Wind sensor		Black box pointing toward the radiometer (i.e. along the boom)? ☒ Y/N		
Temperature / humidity assembly		Fan spins? ☒ Y/N Sounds OK? Y/N		
HygroClip in T/hum assembly		NB: could have a 40 °C offset		
Sonic ranger on AWS		Old sensor with new membrane? ☒ Y/N Lufft replaced		
Sonic ranger on stake(s)		Old sensor with new membrane? ☒ Y/N NB: make sure not being buried next winter "		
Thermistor string		none		
Solar panel		Clean? ☒ Y/N NB: is not effective when partially covered		
GPS antenna		At top of logger enclosure? ☒ Y/N		
Data logger		CR1000 or CR1000X? ☒		
Multiplexer		Clicking during measurement cycle? Y/N N/A		
Logger enclosure		Replaced (including everything inside)? ☒ Y/N New desiccant bags? ☒ Y/N		
Battery box		New box? Y/N New desiccant bags? ☒ Y/N		
Stakes		New one(s) drilled in? Y/N Length of new stake(s) 8 m? Y/N Bottom plugged to reduce melt-in? Y/N		
Tripod		New tripod? Y/N New sensor boom? ☒ Y/N replaced boom + leg that were bent!		

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	more than a day	reset
Download data to PC and/or change CF Card (wait for green light)	Y/N	Y/N
Name of logger program	PROMICE V3 D	
Screen dump of values in fast scan mode	Y/N	Y/N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if tilting towards anemometer)	°	level °
Mast tilt across boom looking from radiometer (- if clockwise)	°	level °
Boom direction relative to north	° true/magn.	good ° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y/N	Y/N
Radiometer exactly aligned with inclinometer? If not, measure.	Y/N	~10 ° Y/N
Temperature / humidity height (bottom of casing – surface) *	260 cm	286 cm
Stake assembly standing?	Y/N / sort of	Y/N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	cm
Free length of stake 2 (if 3: middle)	cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	N/A cm	cm
Vertical difference in surface height (at station – at hose) (approx.)	N/A cm	cm
Length of thermistor string on surface from surface marking	N/A cm	cm
Apparent damage to sensors? If yes, which?	boom Y/N	Y/N

Notes:

* shifted Luft up to rain opening to same as wind



Must-do at next visit:

1. 3 m taller mast / boom